

EDA on HOUSE PRICING DATA

CODE & OUTPUT:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

df = pd.read_csv("House Price Prediction Dataset.csv")
df.head()
```

	Id	Area	Bedrooms	Bathrooms	Floors	YearBuilt	Location	Condition	Garage	Price
0	1	1360	5	4	3	1970	Downtown	Excellent	No	149919
1	2	4272	5	4	3	1958	Downtown	Excellent	No	424998
2	3	3592	2	2	3	1938	Downtown	Good	No	266746
3	4	966	4	2	2	1902	Suburban	Fair	Yes	244020
4	5	4926	1	4	2	1975	Downtown	Fair	Yes	636056

```
df.shape
```

```
df.info()
```

```
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 10 columns):
 #   Column      Non-Null Count  Dtype  
---  --
 0   Id          2000 non-null   int64  
 1   Area        2000 non-null   int64  
 2   Bedrooms    2000 non-null   int64  
 3   Bathrooms   2000 non-null   int64  
 4   Floors       2000 non-null   int64  
 5   YearBuilt    2000 non-null   int64  
 6   Location     2000 non-null   object  
 7   Condition    2000 non-null   object  
 8   Garage       2000 non-null   object  
 9   Price        2000 non-null   int64  
dtypes: int64(7), object(3)
memory usage: 156.4+ KB
```

	Id	Area	Bedrooms	Bathrooms	Floors	YearBuilt	Price
count	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000
mean	1000.500000	2786.209500	3.003500	2.55250	1.993500	1961.446000	537676.855000
std	577.494589	1295.146799	1.424606	1.10899	0.809188	35.926695	276428.845719
min	1.000000	501.000000	1.000000	1.00000	1.000000	1900.000000	50005.000000
25%	500.750000	1653.000000	2.000000	2.00000	1.000000	1930.000000	300098.000000
50%	1000.500000	2833.000000	3.000000	3.00000	2.000000	1961.000000	539254.000000
75%	1500.250000	3887.500000	4.000000	4.00000	3.000000	1993.000000	780086.000000
max	2000.000000	4999.000000	5.000000	4.00000	3.000000	2023.000000	999656.000000

```
df.isnull().sum()
```

```

Id          0
Area        0
Bedrooms    0
Bathrooms   0
Floors       0
YearBuilt   0
Location    0
Condition   0
Garage       0
Price       0
dtype: int64

```

```

df["Location"] = df["Location"].fillna(df["Location"].mode()[0])
df["Condition"] = df["Condition"].fillna(df["Condition"].mode()[0])
df["Garage"] = df["Garage"].fillna(df["Garage"].mode()[0])
le = LabelEncoder()
df['Location'] = le.fit_transform(df['Location'])
df['Condition'] = le.fit_transform(df['Condition'])
df['Garage'] = le.fit_transform(df['Garage'])
df.head()

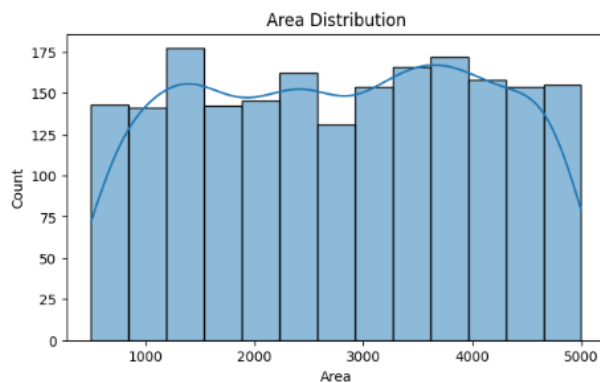
```

	Id	Area	Bedrooms	Bathrooms	Floors	YearBuilt	Location	Condition	Garage	Price
0	1	1360	5	4	3	1970	0	0	0	149919
1	2	4272	5	4	3	1958	0	0	0	424998
2	3	3592	2	2	3	1938	0	2	0	266746
3	4	966	4	2	2	1902	2	1	1	244020
4	5	4926	1	4	2	1975	0	1	1	636056

```

plt.figure(figsize=(7,4))
sns.histplot(df['Area'], kde=True)
plt.title("Area Distribution")
plt.show()

```



```

plt.figure(figsize=(7,4))
sns.histplot(df['Price'], kde=True)

```

```
plt.title("Price Distribution")
```

```
plt.show()
```

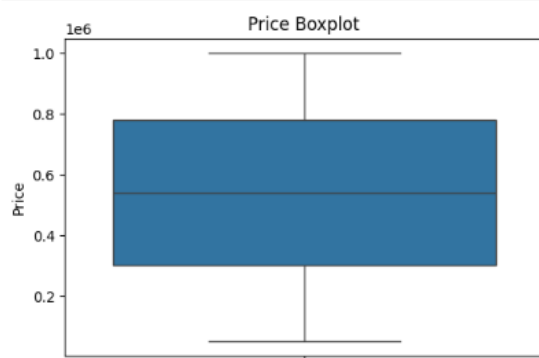


```
plt.figure(figsize=(6,4))
```

```
sns.boxplot(y=df['Price'])
```

```
plt.title("Price Boxplot")
```

```
plt.show()
```

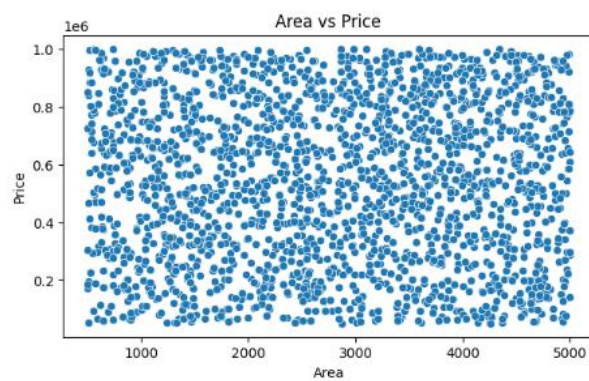


```
plt.figure(figsize=(7,4))
```

```
sns.scatterplot(x='Area', y='Price', data=df)
```

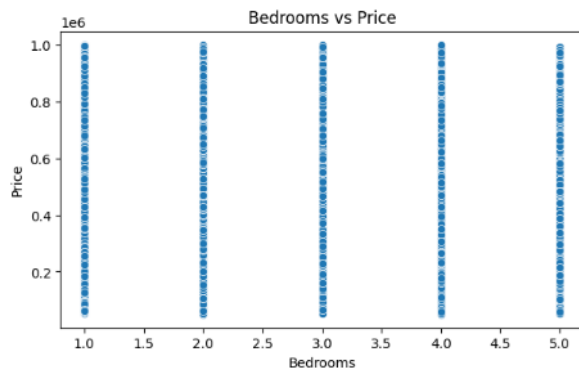
```
plt.title("Area vs Price")
```

```
plt.show()
```

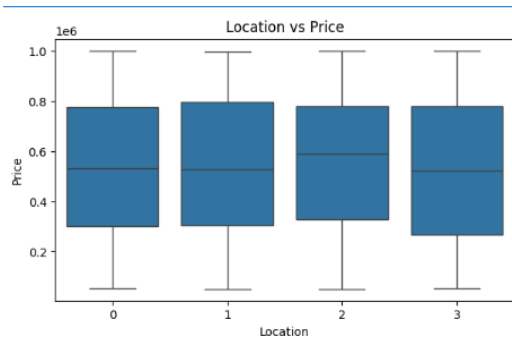


```
plt.figure(figsize=(7,4))
```

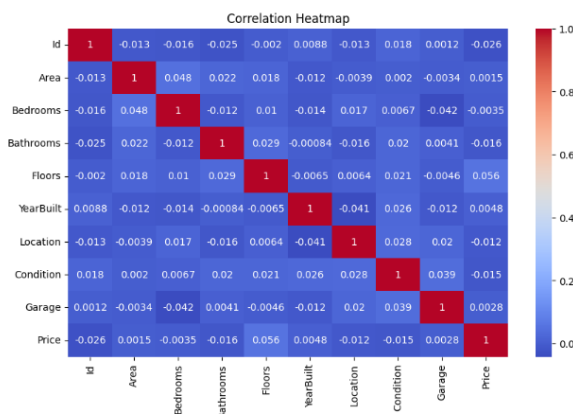
```
sns.scatterplot(x='Bedrooms', y='Price', data=df)
plt.title("Bedrooms vs Price")
plt.show()
```



```
plt.figure(figsize=(7,4))
sns.boxplot(x='Location', y='Price', data=df)
plt.title("Location vs Price")
plt.show()
```

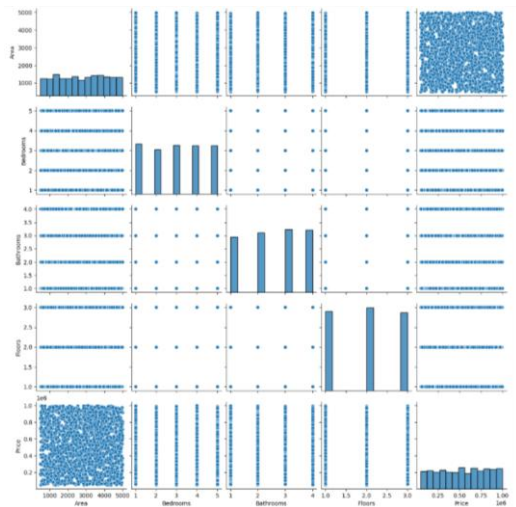


```
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



```
sns.pairplot(df[['Area', 'Bedrooms', 'Bathrooms', 'Floors', 'Price']])
```

```
plt.show()
```



```
X = df[['Area','Bedrooms','Bathrooms','Floors','Location','Condition','Garage']]
```

```
y = df['Price']
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.2, random_state=42)
```

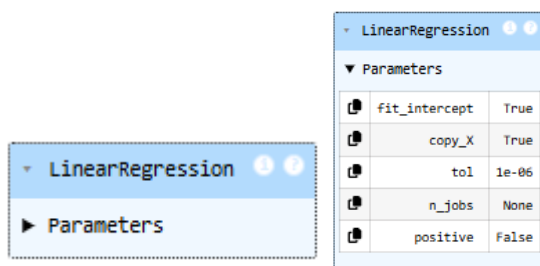
```
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
```

```
X_test = scaler.transform(X_test)
```

```
model = LinearRegression()
```

```
model.fit(X_train, y_train)
```



```
y_pred = model.predict(X_test)
```

```
print("MAE :", mean_absolute_error(y_test, y_pred))
```

```
print("MSE:", mean_squared_error(y_test, y_pred))
```

```
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
```

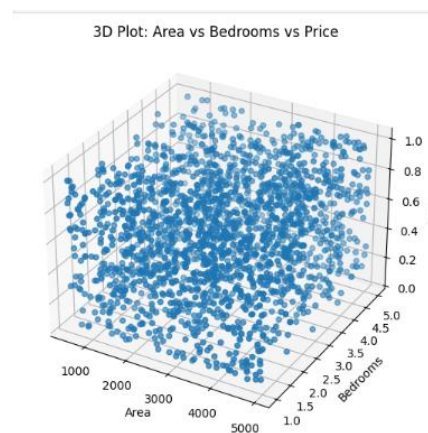
```
print("R² :", r2_score(y_test, y_pred))
```

```
MAE : 244206.5732734258
MSE : 78640911107.24205
RMSE: 280429.8684292421
R² : -0.01082384662292557
```

```

from mpl_toolkits.mplot3d import Axes3D
fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')
ax.scatter(df['Area'], df['Bedrooms'], df['Price'])
ax.set_xlabel("Area")
ax.set_ylabel("Bedrooms")
ax.set_zlabel("Price")
ax.set_title("3D Plot: Area vs Bedrooms vs Price")
plt.show()

```



```
pd.DataFrame(model.coef_, X.columns, columns=['Coefficient'])
```

Coefficient	
Area	-570.470294
Bedrooms	161.247249
Bathrooms	-11061.225727
Floors	19057.657909
Location	-3091.386269
Condition	-2768.759930
Garage	1859.738687